Yuan Jiang

Ph.D. Candidate, Industrial & Enterprise Systems Engineering
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My research lies at the intersection of physics-informed AI, multi-fidelity simulation, and digital twin technologies: a convergence reshaping how we model, monitor, and sustain complex engineering systems. I develop principled frameworks that integrate governing physics with data-driven learning to enable reliability-aware design, scalable health monitoring, and adaptive life-cycle decision-making under data-scarce, safety-critical conditions, with applications in battery thermal management, aerospace electric propulsion, and multi-physics mechatronic systems.

Education

University of Illinois Urbana-Champaign **Urbana, IL, USA**
Ph.D., Industrial Engineering (Minor: Statistics) Aug 2023 – May 2027 (Expected)
Advisor: Prof. Pingfeng Wang
Dissertation: Physics-informed multi-fidelity digital twins for design, diagnosis, and prognosis of complex engineering systems

Tongji University **Shanghai, China**
M.Sc., Vehicle Engineering Sep 2020 – Jun 2023
Advisor: Prof. Gang Niu
Thesis: Time-frequency feature extraction and health monitoring of rail short-pitch damage

Tongji University **Shanghai, China**
B.Sc., Vehicle Engineering Sep 2015 – Jul 2020

Industrial Experience

Rivian Automotive, Inc. **Champaign, IL, USA**
Computational Methods Development Intern May – Aug 2026
Developed a unified multi-objective optimization framework for electric motor design, integrating heuristic and gradient-based solvers with trust-region surrogate modeling strategies to reduce computational cost while maintaining design fidelity

NIO Automobile, Inc. **Shanghai, China**
Power Systems Algorithm Intern May – Aug 2023
Developed data-driven condition monitoring and fault detection methods for EV power swap and charging systems, translating large-scale operational and sensor data into real-time health indicators and early-warning signals

HiRain Technologies **Shanghai, China**
Autonomous Driving Development Intern Jun – Aug 2022
Contributed to perception software development and data processing pipelines for multi-camera autonomous driving systems, with exposure to large-scale, high-throughput data integration and real-time system constraints

NIO Automobile, Inc. **Shanghai, China**
Battery Testing Intern Feb – May 2021
Conducted experimental testing and data analysis for lithium-ion batteries, supporting model validation and data-informed performance assessment under real-world operating conditions

Awards & Honors

Research Awards

1. **ASME DAC Johnson Controls Best Paper Award**, *Design Automation Committee (DAC), ASME* 2026
2. **IISE QCRE Best Student Poster Award**, *IISE Annual Conference & Expo* 2026
3. **IISE M&D Best Track Paper Finalist**, *IISE Annual Conference & Expo* 2026
4. **Best Poster Award (Second Place)**, *ISE Student Conference, UIUC* 2026
5. **ASME DAC Paper of Distinction**, *Design Automation Committee (DAC), ASME* 2025
6. **ASME IMECE Best Student Paper Award**, *ASME* 2024
7. **Outstanding Master's Thesis Award**, *Tongji University* 2023
8. **IEEE PHM Best Paper Award**, *IEEE Reliability Society* 2022

Fellowships, Scholarships & Academic Honors

1. **Mavis Future Faculty Fellow**, *University of Illinois Urbana-Champaign* 2026
2. **Jerry S. Dobrovolny Fellowship**, *University of Illinois Urbana-Champaign* 2023
3. **Tongji Outstanding Graduate**, *Tongji University* 2023
4. **National Scholarship of China**, *Ministry of Education of the People's Republic of China* 2019, 2022
5. **Shanghai Outstanding Graduate**, *Shanghai Municipal Education Commission* 2020
6. **First-Class Scholarship**, *Tongji University* 2016, 2017, 2018

Professional Recognition

1. **IOE-ISyE-MS&E Joint Rising Star**, *IOE (UMich), ISyE (Georgia Tech), MS&E (Stanford)* 2026
2. **PHM Society Doctoral Symposium**, *PHM Society Annual Conference* 2026
3. **IEEE TIM Outstanding Reviewer**, *IEEE Instrumentation and Measurement Society* 2025

Research Experience

Physics-Guided Multi-fidelity Digital Twins for Hydraulic Drivetrain Remanufacturing Diagnostics
Sponsor: Department of Energy (DOE) REMADE Institute, National Science Foundation (NSF)
Collaboration: University of Connecticut, John Deere

- **Scope:** This project aims to develop a digital twin platform for agricultural hydraulic power transmission systems that integrates physics-based simulation, multi-fidelity modeling, physics-informed machine learning, and inverse diagnosis to support condition monitoring, fault identification, and remanufacturing decision-making for drivetrain components subject to structural defects under diverse operating conditions
- **Multi-Fidelity Simulation:** Developed and refined fluid-structure-contact interaction (FSCI) simulation models in COMSOL with moving mesh and adaptive remeshing; designed a reduced-fidelity simulation strategy by substituting explicit contact pairs with spring-damper approximations, reducing simulation cost to one-quarter of high-fidelity baseline while preserving dominant transient pressure dynamics
- **Physics-Guided Neural Operator:** Proposed a physics-guided Fourier deep compositional neural operator with Fourier feature embeddings and two-stage multi-fidelity training strategy for transient hydraulic pressure prediction under clutch wear and missing-plate defect conditions with limited high-fidelity data, achieving pressure prediction errors within 2% and reducing extrapolation error by 10% compared to neural operator baselines
- **Digital-Twin-Based Diagnosis:** Proposed an inverse diagnosis framework by constructing condition-labeled reference libraries from the trained model and performing defect identification via time-series similarity matching, achieving 98% diagnostic accuracy within 1 s; mentored one junior Ph.D. student on comparative evaluation of similarity measures under realistic noise conditions
- **Outcomes & Impact:** Submitted **1 journal paper** (co-first-authored) under second-round review in ASME JMD and **1 peer-reviewed conference paper** (co-first-authored) accepted at ASME IDETC 2026; ongoing extensions to trustworthy diagnosis with confidence quantification and post-remanufacturing performance prediction

Physics-Informed Adaptive Digital Twin for Hall Thruster Erosion Degradation Modeling and Prognostics
Sponsor: Air Force Office of Scientific Research (AFOSR)
Collaboration: UIUC Aerospace Engineering, Georgia Institute of Technology

- **Scope:** This project aims to develop an intelligent health monitoring and remaining useful life (RUL) assessment digital twin for Hall thruster electric propulsion systems, where direct in-situ measurement of channel wall erosion is infeasible and degradation patterns vary across thruster units; the framework enables uncertainty-aware prognostics that provide prior information for refueling and maintenance decision-making in cislunar space mission logistics
- **Degradation Mechanism Discovery:** Built a 2D particle-in-cell (PIC) plasma simulation environment in VSim and extended the HallThruster.jl open-source 1D fluid-based plasma solver with an erosion profile estimation loop; proposed a multi-fidelity calibration strategy that integrates 2D ion flux ratios into 1D solver, improving erosion profile accuracy by 10% while reducing lifecycle simulation from 2 weeks to 6 hours; generated lifecycle erosion degradation datasets and discovered trust force as an observable indicator for in-situ prognostics
- **Adaptive Digital Twin Update:** Proposed an adaptive self-cognizant dynamic system (ASDS) that formulates erosion degradation as a mechanism-informed state-space system with neural networks serving as state transition and measurement modelers, trained offline on multi-fidelity simulation data; during online monitoring, particle filtering estimates state-of-health and updates network parameters by assimilating in-flight telemetry, achieving estimation error within 5% with per-step update time of 0.15 s for real-time model adaptation under varying operational conditions
- **Uncertainty-Aware Prognostics:** Proposed uncertainty-aware RUL prediction methodology considering the manufacturing variability and environmental disturbances on degradation patterns; achieved RUL prediction error within 5% over a 5000-h lifecycle case study on the SPT-100 thruster, improving prediction accuracy by over 3% compared to baseline methods

- **Outcomes & Impact:** Published **1 journal paper** (first-authored) in ASME JMD (invited) and **2 peer-reviewed conference papers** (first-authored) at RAMS 2025 and IISE 2026; received **ASME DAC Paper of Distinction** (Top 10 out of 103) and **IISE QCRE Best Student Poster Award** (1st out of 18)

Physics-Informed Machine Learning for Large-Scale Battery Thermal-Health Management and Design
Sponsor: National Science Foundation (NSF)
Collaboration: Stanford University, UIUC Mechanical Engineering, John Deere, Ford

- **Scope:** This project aims to develop physics-informed machine learning digital twins for battery pack thermal management and health monitoring, enabling real-time layout-aware temperature field prediction for design under limited high-fidelity data, coupled with multi-modal sensing and data fusion for scalable condition monitoring and anomaly detection in large-scale battery systems
- **Multi-Fidelity Physics-Informed CNN:** Developed multi-fidelity thermal simulation models in COMSOL for indirect liquid cooling battery packs, simplifying 3D conjugate heat transfer into an experimentally validated equivalent 2D representation with 6× speedup; proposed a multi-fidelity physics-informed convolutional neural network (MFPI-CNN) that embeds heat transfer PDEs and boundary conditions into an unsupervised pre-training backbone, and fine-tunes with limited high-fidelity data, achieving 2% prediction error while reducing labeled data requirements to one-fifth of purely data-driven baselines on an 8-cell battery pack
- **Physics-Informed Neural Operator:** Proposed a series of mesh-free physics-informed neural operators, from cross-attention operator network (PI-CAON) to self-cross attention operator network (PI-SCAON), incorporating Fourier feature encoding for multi-frequency thermal patterns, hard-constrained boundary enforcement, and stacked attention mechanisms to capture inter-cell thermal interactions in large-scale battery packs; achieved prediction errors within 2% on a 20-cell battery pack using data-free training, with inference speed 250× faster than finite element simulation
- **Design Optimization & Thermal Safety:** Formulated a surrogate-based battery layout optimization framework coupling physics-informed models with heuristic algorithms, achieving 7.2% improvement in temperature uniformity; extended the framework to thermal runaway propagation prediction using Fourier neural operator (FNO) and developed local-to-global inference strategies enabling zero-shot deployment from small-scale to large-scale battery packs without retraining
- **Multi-modal Sensing for Health Monitoring:** Built a physics-based electromagnetic simulation framework to model busbar current-magnetic field relationships, enabling inverse reconstruction of cell-level currents from magnetoresistive (MR) sensor readings; developed a sensor placement optimization strategy based on proper orthogonal decomposition (POD) to maximize observability, reducing required sensors from 9 to 4 out of 500 candidate locations; experimental validation on a 9-cell battery pack in collaboration with Stanford University
- **Outcomes & Impact:** Published **2 journal papers** (1 first-authored and 1 co-first-authored) in RESS and EAAI, and **5 peer-reviewed conference papers** (1 first-authored and 2 co-first-authored) at ASME IDETC 2026, IISE 2026, RAMS 2025, and IEEE ITEC 2025; received **ASME DAC Best Paper Award** (1st out of 107) and **IISE M&D Best Track Paper Finalist** (Top 4 out of 40)

Signal Processing and Digital Twin for Structural and Mechatronic System Health Monitoring
Sponsor: National Science Foundation (NSF)

- **Scope:** This project aims to develop advanced time-frequency signal processing methodologies and simulation-based digital twins for diagnostics and health monitoring of structural and mechatronic systems operating under variable operating conditions, where conventional methods fail due to non-stationary dynamics, multimodal signal overlap, and noise interference
- **Damage Mechanism Discovery:** Constructed a vehicle-track multi-body dynamics model in Simpack to simulate axle-box vibration responses under various rail damage scenarios; discovered the quantitative relationships between damage severity, train speed, and axle-box vibration characteristics, establishing physics-based features for rail health assessment; proposed a rail impact index for damage localization, achieving 98% localization accuracy on field data from operational tram lines
- **Time-Frequency Signal Decomposition:** Proposed a family of iterative multimodal signal decomposition algorithms (IFETF, IAVKF, IDVKF) for damage feature extraction in structural health monitoring and non-stationary fault detection in mechatronic transmissions under variable speed conditions; improved damage feature detection performance by 5–20 dB; validated on Lamb wave inspection, rail damage detection, gearbox tooth crack detection, and bearing fault diagnosis
- **Outcomes & impact:** Published **5 journal papers** (3 first-authored) in MSSP, IEEE TII, and IEEE IoTJ, and **3 peer-reviewed conference papers** (2 first-authored) in PHM 2021 and 2022; received **IEEE PHM Best Paper Award** (Top 5 out of 274) and **Outstanding Master's Thesis Award** (Top 5%)

Teaching & Mentoring Experience

Teaching interests include reliability engineering, signal processing, engineering design analysis, design optimization, and decision-making under uncertainty. At the graduate level, I am interested in physics-informed machine learning, digital twin modeling, and data-driven design optimization for safety-critical engineering systems.

Teaching Experience

SE 311: Engineering Design Analysis **University of Illinois Urbana-Champaign**
Main instructor: Prof. Krishnan Girish Spring 2026

- Delivered guest lectures on stress analysis, covering state of stress, stress transformation, principal stresses, and maximum shear stress, with emphasis on engineering applications and analytical problem solving
- Delivered exam review lectures covering elasticity, stress analysis, energy methods, and introductory finite element analysis, reinforcing systematic approaches to solving engineering mechanics problems
- Developed and delivered a virtual Abaqus tutorial for finite element analysis, guiding students through finite element modeling, post-processing, and project completion involving stress analysis and validation against analytical solutions

SE 411: Reliability Engineering **University of Illinois Urbana-Champaign**
Main instructor: Prof. Krishnan Girish Spring 2025

- Delivered guest lectures on reliability analysis and reliability-based design optimization (RBDO), covering problem formulation, algorithmic implementation, and engineering interpretation.
- Designed an RBDO course project and mentored students on modeling, optimization algorithms, and coding implementation
- Led exam review sessions with structured problem-solving tutorials to reinforce core concepts and analytical skills

SE 450: Decision Analysis **University of Illinois Urbana-Champaign**
Main instructor: Prof. Pingfeng Wang Fall 2024, Fall 2025

- Contributed to undergraduate instruction by leading discussion sections and providing academic support via office hours
- Delivered exam review lectures covering multivariate decision making, decision trees, and uncertainty-based decision making, with an emphasis on systematic problem formulation and solution strategies

Rail Vehicle Systems Health Management **Tongji University**
Main instructor: Prof. Gang Niu (Instructed in English) Spring 2022

- Supported course delivery by assisting lecturers and coordinating laboratory components for an undergraduate course
- Led laboratory sessions on bearing and gearbox test rigs, covering experimental setup, data acquisition, and signal processing
- Instructed students on vibration-based fault diagnosis using FFT, wavelet transforms, and machine learning methods

Student Mentoring

Jiayi Zhao *(Ph.D. Student, UIUC)* **Jan 2026 – Present**
Digital-Twin-Based Inverse Diagnosis for Hydraulic Drivetrain Remanufacturing

- Primary technical advisor for research methodology, guiding digital-twin framework design, similarity-based diagnostic approach, and trustworthy decision-making methods.
- Co-first-authored ASME IDETC 2026 conference paper (accepted); journal paper on trustworthy diagnosis in preparation.

Jingsen Yang *(M.S., UIUC)* **Nov 2025 – Present**
Multi-fidelity Surrogate Modeling and Data Fusion based on Foundational Tabular Model

- Primary technical advisor for surrogate modeling, data processing, and methodological development.
- Thesis and conference paper in preparation.

Yanbin Luo *(M.S., UIUC)* **Jul 2025 – May 2026**
Experimental Degradation Modeling of High-Electron-Mobility Transistors (HEMTs) under Extreme Operating Conditions

- Primary technical advisor for experimental design, data processing, and degradation modeling.
- Guided master's thesis and IISE conference paper preparation (accepted and presented).

Academic Service

Conference Leadership

- Session Chair, Prognostics and Reliability in Power and Energy Systems, *IISE Annual Conference & Expo* 2026

Departmental & University Service

- ISE Graduate Program Ambassador, *Industrial & Enterprise Systems Engineering, UIUC* 2026 – Present

Journal Reviewer

- Reliability Engineering & System Safety
- Engineering Applications of Artificial Intelligence
- Expert Systems with Applications
- IEEE Transactions on Reliability
- Neurocomputing
- Journal of Process Control
- Journal of Vibration and Control
- Scientific Reports
- IEEE Transactions on Industrial Informatics
- Mechanical Systems and Signal Processing
- IEEE Transactions on Instrumentation & Measurement
- ISA Transactions
- IEEE Sensors Journal
- Computers & Structures
- Microelectronics Journal
- Journal of Rail and Rapid Transit

Conference Reviewer

- ASME International Design Engineering Technical Conferences & Computers and Information in Engineering Conference (IDETC-CIE)

Professional Memberships

- Institute of Electrical and Electronics Engineers (IEEE)
- American Society of Mechanical Engineers (ASME)
- Institute of Industrial and Systems Engineers (IISE)
- Institute for Operations Research and the Management Sciences (INFORMS)

Publications

[#] indicates equal contribution

11 published peer-reviewed journal papers (6 first-author), including IEEE TII, RESS, MSSP, EAAI, and ASME JMD.

Journal Papers

[J1] **Yuan Jiang**, Yiyue Jiang, Zheng Liu, and Pingfeng Wang. Physics-informed cross-attention operator network with hard-constrained Fourier features for heat map prediction of large-scale battery packs. *Engineering Applications of Artificial Intelligence*, 176 (2026) 114833. DOI: 10.1016/j.engappai.2026.114833.

- ISE Student Conference Best Poster Award Second Place (2nd out of 28)

[J2] **Yuan Jiang**, Yiyue Jiang, Zheng Liu, Yuejian Chen, and Pingfeng Wang. Iterative dispersive Vold-Kalman filter with local adaptive bandwidth for dispersive signal decomposition in structural health monitoring. *IEEE Transactions on Industrial Informatics*, (2026). DOI: 10.1109/TII.2026.3702939.

[J3] Yiyue Jiang, **Yuan Jiang**, and Pingfeng Wang. A review on mitigating thermal runaway propagation in battery packs: From mechanisms to modeling and design optimization. *Energy Advances*, 5(6) (2026) 777-809. DOI: 10.1039/D5YA00351B.

[J4] **Yuan Jiang**, Alexandra N. Leeming, Joshua L. Rovey, and Pingfeng Wang. Remaining useful life prediction for Hall thrusters based on adaptive self-cognizant dynamic system and multi-physics modeling. *Journal of Mechanical Design*, 148(7) (2026) 071701. DOI: 10.1115/1.4070967.

- ASME DAC Paper of Distinction (Top 10 out of 103)

[J5] Zheng Liu, Yanwen Xu, **Yuan Jiang**, Anabel Renteria, Parth Bansal, Chenlong Xu, Pingfeng Wang, and Yumeng Li. Uncertainty quantification of additively manufactured architected cellular materials for energy absorption applications. *ASCE-ASME Journal of Risk and Uncertainty in Engineering Systems, Part B: Mechanical Engineering*, 11(3) (2025). DOI: 10.1115/1.4066933.

[J6] Pouya Kabirzadeh, Zheng Liu, Mostafa Olyaei, Haoyun Qiu, Yashraj Gurumukhi, Harsh Tyagi, **Yuan Jiang**, Vivek S Garimella, Bakhshish Preet Singh, Yumeng Li, Pingfeng Wang, and Nenad Miljkovic. Integrating heat transfer and control optimization: A comprehensive review of battery thermal management systems. *Journal of Energy Storage*, 131 (2025) 117289. DOI: 10.1016/j.est.2025.117289.

[J7] **Yuan Jiang**[#], Zheng Liu[#], Pouya Kabirzadeh, Yulun Wu, Yumeng Li, Nenad Miljkovic, and Pingfeng Wang. Multi-fidelity physics-informed convolutional neural network for heat map prediction of battery packs. *Reliability Engineering & System Safety*, 256 (2025) 110752. DOI: 10.1016/j.ress.2024.110752.

[J8] Yuejian Chen, Zihan Li, **Yuan Jiang**, Chunsheng Yang, Min Xia, and Ke Feng. Enhanced sparse LPV-ARMA model with ensemble basis functions for mechatronic transmission fault detection under variable speed conditions. *IEEE Internet of Things Journal*, 12(11) (2025) 17223-17232. DOI: 10.1109/JIOT.2025.3535580.

[J9] Yuejian Chen, Zihan Li, **Yuan Jiang**, Dao Gong, and Kai Zhou. Sparse LPV-ARMA model for non-stationary vibration representation and its application on gearbox tooth crack detection under variable speed conditions. *Mechanical Systems and Signal Processing*, 224 (2025) 112161. DOI: 10.1016/j.ymssp.2024.112161.

[J10] **Yuan Jiang**, Yuejian Chen, and Pingfeng Wang. An iterative adaptive Vold-Kalman filter for non-stationary signal decomposition in mechatronic transmission fault diagnosis under variable speed conditions. *IEEE Transactions on Industrial Informatics*, 20(8) (2024) 10510-10519. DOI: 10.1109/TII.2024.3393536.

[J11] **Yuan Jiang** and Gang Niu. An iterative frequency-domain envelope-tracking filter for dispersive signal decomposition in structural health monitoring. *Mechanical Systems and Signal Processing*, 179 (2022) 109329. DOI: 10.1016/j.ymssp.2022.109329.

Journal Papers under Review

[J12] **Yuan Jiang**[#], Parth Bansal[#], Mohammad Mundiwala, Chao Hu, and Pingfeng Wang. Physics-guided multi-fidelity neural operator digital twin for wet clutch hydraulic pressure prediction. *Journal of Mechanical Design.* (Under second round of review)

Journal Papers in Preparation

[J13] Yulun Wu, **Yuan Jiang**, Yumeng Li, and Pingfeng Wang. From local to global without retraining: Boundary-aware inference strategies for fixed-size surrogates on larger domains in battery temperature field prediction. *To be submitted to Journal of Energy Storage by Sep. 2026.*

[J14] Yiyue Jiang, **Yuan Jiang**, and Pingfeng Wang. Onset-aware spatiotemporal forecasting of temperature fields in battery thermal runaway propagation. *To be submitted to Applied Energy by Sep. 2026.*

[J15] **Yuan Jiang**, Yiyue Jiang, Yanwen Xu, and Pingfeng Wang. Adaptive surrogate modeling and uncertainty quantification for battery pack thermal runaway based on operator learning and conformal prediction. *To be submitted to Applied Energy by Nov. 2026.*

[J16] Jiayi Zhao[#], **Yuan Jiang**[#], Chao Hu, and Pingfeng Wang. Trustworthy remanufacturing diagnosis for hydraulic drivetrain based on physics-guided multi-fidelity digital twins. *To be submitted to IEEE Transactions on Reliability by Nov. 2026.*

[J17] **Yuan Jiang**, Yiming Zhao, Debbie Senesky, and Pingfeng Wang. Optimal magnetic sensor placement for battery pack cell-level condition monitoring based on physics-informed digital twin. *To be submitted to IEEE Transactions in Industrial Electronics by Jan. 2027.*

[J18] **Yuan Jiang**, Xiao Huan, Ying Diao, and Pingfeng Wang. Geometry-aware neural operators for strain-rate field prediction and nozzle design in direct ink 3D printing. *To be submitted to Additive Manufacturing by March 2027.*

Conference Proceedings

[C1] **Yuan Jiang** and Pingfeng Wang. Physics-informed self-cross attention operator network for heat map prediction in large-scale battery thermal management systems. *ASME International Engineering Technical Conferences & Computers and Information in Engineering Conference (IDETC 2026)*, Houston, TX, USA. 24 Aug. 2026, 10 pages. (Accepted)

- ASME DAC Johnson Controls Best Paper Award (1st out of 107)

[C2] Jiayi Zhao[#], **Yuan Jiang**[#], Yumeng Li, and Pingfeng Wang. A digital-twin-based framework for inverse diagnosis of wet clutch hydraulic pressure signals using time-series similarity matching. *ASME International Engineering Technical Conferences & Computers and Information in Engineering Conference (IDETC 2026)*, Houston, TX, USA. 24 Aug. 2026, 10 pages. (Accepted)

[C3] Yulun Wu, **Yuan Jiang**, Yumeng Li, and Pingfeng Wang. From local to global: CNN surrogates for scaling layout-resolved temperature prediction in battery thermal management. *ASME International Engineering Technical Conferences & Computers and Information in Engineering Conference (IDETC 2026)*, Houston, TX, USA. 24 Aug. 2026, 10 pages. (Accepted)

[C4] **Yuan Jiang** and Pingfeng Wang. Multi-fidelity adaptive self-cognizant dynamic system for remaining useful life prediction of Hall thrusters. *IISE Annual Conference & Expo (IISE 2026)*, Arlington, TX, USA. 17 May 2026, 6 Pages. (Accepted and Presented)

- IISE QCRE Best Student Poster Award (1st out of 18)

[C5] Yiyue Jiang[#], **Yuan Jiang**[#], and Pingfeng Wang. Spatiotemporal design-aware surrogate modeling of thermal runaway propagation in Li-ion battery packs. *IISE Annual Conference & Expo (IISE 2026)*, Arlington, TX, USA. 17 May 2026, 6 Pages. (Accepted and Presented)

- IISE M&D Best Track Paper Finalist (Top 4 out of 40)

[C6] Yanbin Luo, **Yuan Jiang**, and Pingfeng Wang. Degradation behavior and health monitoring of GaN HEMTs under high-temperature stress conditions. *IISE Annual Conference & Expo (IISE 2026)*, Arlington, TX, USA. 17 May 2026, 6 Pages. (Accepted and Presented)

[C7] Jiayi Zhao, **Yuan Jiang**, Yiyue Jiang, and Pingfeng Wang. A complexity-aligned benchmark for adaptive sampling in GP global fitting. *IISE Annual Conference & Expo (IISE 2026)*, Arlington, TX, USA. 17 May 2026, 6 Pages. (Accepted and Presented)

[C8] Alexandra N. Leeming, **Yuan Jiang**, Stephen Messing, Siyuan Wang, Pingfeng Wang, J. Gary Eden, Peter D. Dragic, Joshua L. Rovey, Valentin Korman, and Kurt Polzin, Exploration of an in-situ Hall thruster erosion and plasma diagnostic sensor. *AIAA Science and Technology Forum and Exposition (AIAA SciTech 2026)*, Orlando, FL, USA. 12 Jan. 2026, 14

Pages. DOI: 10.2514/6.2026-0929.

[C9] Zheng Liu[#], **Yuan Jiang**[#], Yumeng Li, and Pingfeng Wang, Physics-informed machine learning enhanced battery pack optimization. *IEEE/AIAA Transportation Electrification Conference and Electric Aircraft Technologies Symposium (ITEC+EATS 2025)*, Anaheim, CA, USA. 18 June 2025, 5 Pages. DOI: 10.1109/ITEC63604.2025.11098093.

[C10] **Yuan Jiang**, Alexandra N. Leeming, Joshua L. Rovey, and Pingfeng Wang, Prognostics of Hall thruster erosion using multiphysics-based modeling and machine learning. *Annual Reliability and Maintainability Symposium (RAMS 2025)*, Destin, FL, USA. 27 Jan. 2025, 7 Pages. DOI: 10.1109/RAMS48127.2025.10935282.

[C11] Zheng Liu, **Yuan Jiang**, Yumeng Li, and Pingfeng Wang, Physics-informed machine learning for battery pack thermal management. *Annual Reliability and Maintainability Symposium (RAMS 2025)*, Destin, FL, USA. 27 Jan. 2025, 7 Pages. DOI: 10.1109/RAMS48127.2025.10935157.

[C12] Shimeng Yang, **Yuan Jiang**, Zheng Liu, and Pingfeng Wang, Transductive transfer learning features for prognostics and health management. *Annual Reliability and Maintainability Symposium (RAMS 2025)*, Destin, FL, USA. 27 Jan. 2025, 7 Pages. DOI: 10.1109/RAMS48127.2025.10935143.

[C13] Parth Bansal, **Yuan Jiang**, Zhou Li, Sergio Cordero, Zahra Heussen, Debbie Senesky, Pingfeng Wang, and Yumeng Li, Multiphysics modeling and simulation of gas sensor for NO_2 detection. *ASME International Mechanical Engineering Congress and Exposition (IMECE 2024)*, Portland, OR, USA, 17 Nov. 2024, 6 Pages. DOI: 10.1115/IMECE2024-145663.

- ASME IMECE Best Student Paper Award

[C14] **Yuan Jiang** and Gang Niu, Rail local damage detection based on recursive frequency-domain envelope tracking filter and rail impact index. *IEEE Global Reliability and Prognostics and Health Management Conference (PHM 2022)*, Yantai, China. 13 Oct. 2022, 7 Pages. DOI: 10.1109/PHM-Yantai55411.2022.9942082.

- IEEE PHM Best Paper Award (Top 5 out of 274)

[C15] Hongyang Zhao, **Yuan Jiang**, and Gang Niu, A tacholess order tracking method based on extended intrinsic chirp component decomposition for gears under large speed variation conditions. *Journal of Physics: Conference Series*, 2184(1) (2022) 12052. DOI: 10.1088/1742-6596/2184/1/012052.

[C16] **Yuan Jiang**, Hongyang Zhao, and Gang Niu, Intelligent rolling bearing fault diagnosis under variable speed conditions without tachometers. *IEEE Global Reliability and Prognostics and Health Management Conference (PHM 2021)*, Nanjing, China. 15 Oct. 2021, 7 Pages. DOI: 10.1109/PHM-Nanjing52125.2021.9613049.